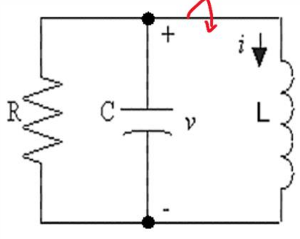
Initial conditions-2

Example: Consider the parallel RLC circuit with R = 604 mΩ, L = 20 mH and C = 207 mF
Note that these values are approximate, but come close to giving the correct values for s = -4 +/- j15



$$s = -4 \pm j15$$
$$i_n = e^{-4t}\left(B_1 \cos 15t + B_2 \sin 15t\right)$$
$$i(0) = 0\text{ A}$$
$$v(0) = 30\text{ V}$$

We want to find the values of B_1 and B_2. Let's apply the first initial condition by evaluating i_n from the solution form and equating this to the value of the initial condition (IC):

$$i_n(0) = e^{-4\cdot 0}\left(B_1 \cos 15\cdot 0 + B_2 \sin 15\cdot 0\right) = 0$$
$$1\cdot\left(B_1\cdot 1 + B_2 \cdot 0\right) = 0$$
$$B_1 = 0$$

Good, that gets rid of half the solution. But, how to use v(0)? Well

Find B2?

Initial conditions-3

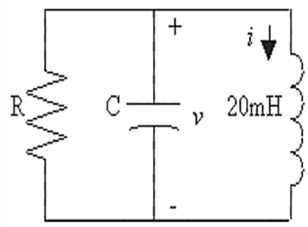
$$v = L\frac{di}{dt} \quad \text{so} \quad \frac{di}{dt}(0) = \frac{v(0)}{L} = \frac{30\text{ V}}{20\text{ mH}} = 1500\text{ A/s}$$

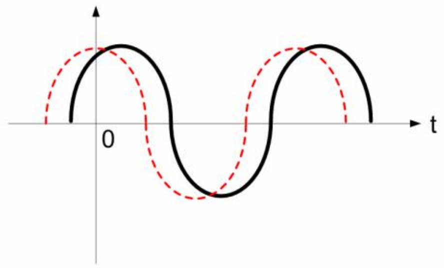
To get $\frac{di}{dt}(0)$ from the solution form, we have to differentiate it. It's a good thing $B1$ was zero, it makes the differentiation half the effort.

$$\frac{di_n}{dt} = \frac{d}{dt}e^{-4t}\left(B_2 \sin 15t\right) = -4e^{-4t}\left(B_2 \sin 15t\right) + e^{-4t}\left(15\cdot B_2 \cos 15t\right)$$

$$\frac{di_n}{dt}(0) = -4\cdot 1\cdot\left(B_2\cdot 0\right) + 1\cdot\left(15\cdot B_2 \cdot 1\right) = 1500$$

$$B_2 = \frac{1500}{15} = 100 \quad \text{and} \quad i_n = 100e^{-4t}\sin 15t$$





$-\ V_m \cos(\omega t)$

$--\ V_m \cos(\omega t + \phi)$

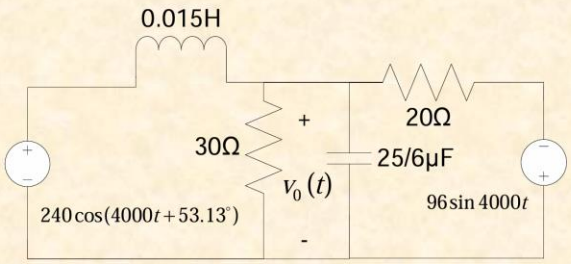
ϕ: phase advance

move waveform to the left

ϕ is the phase angle of the waveform and its reference to where the waveform begins

SOURCE TRANSFORMATION AND THEVENIN-NORTON EQUIVALENT CIRCUIT

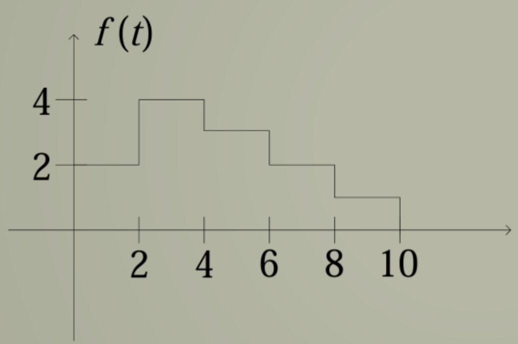
Example:



Find $v_0(t) = ?$

左右两个部分分别使用戴维南等效，再分析节点的电流，使用基尔霍夫定律

Example:



more complex functions can be decomposed into a series (sum) of unit step

$$f(t) = 2u(t) + 2u(t-2) - u(t-4)$$
$$-u(t-6) - u(t-8) - u(t-10)$$

Laplace Transform /Operational Transform

Integration:

Using integral by parts $\int u\,dv = uv - \int v\,du$

$$L\left\{\int_{0^-}^{t} f(x)\,dx\right\} = \int_{0^-}^{\infty}\left[\int_{0^-}^{t} f(x)\,dx\right]e^{-st}dt$$

We can do it by integration by parts or using Laplace Transform Derivative results.

$$u = \int_{0^-}^{t} f(x)\,dx;\ dv = e^{-st}dt;\ du = f(t)\,dt;\ v = -\frac{e^{-st}}{s}\Big|_{0^-}^{\infty}$$

w.r.t. to time, t

$$\left[-\frac{e^{-st}}{s}\int_{0^-}^{t} f(x)\,dx\right]_{0^-}^{\infty} + \int_{0^-}^{\infty}\frac{e^{-st}}{s} f(t)\,dt = 0 + \frac{F(s)}{s} = \frac{F(s)}{s}$$

Alternate proof: $L\left\{\int_{0^-}^{t} f(x)\,dx\right\} = L\{g(t)\};\ \frac{dg}{dt} = f(t);\ g(0) = 0$

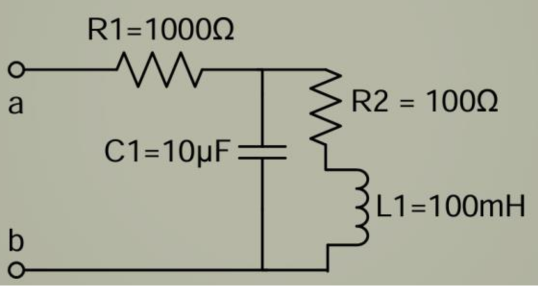
$$L\left\{\frac{d}{dt}f(t)\right\} = sF(s) - f(0^-)$$

$$L\left\{\frac{dg}{dt}\right\} = L\{f(t)\};\ sG(s) = F(s);\ G(s) = \frac{F(s)}{s}$$

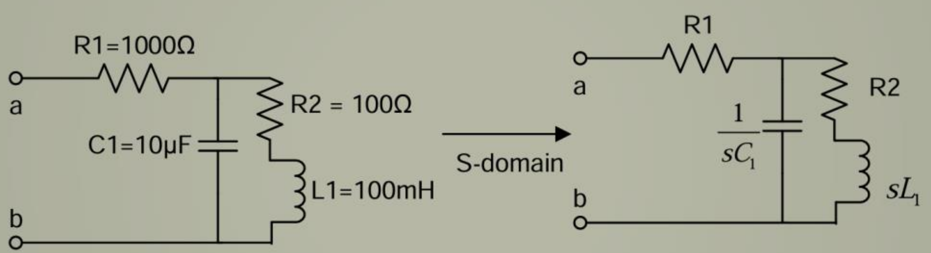
$$G(s) = L\left\{\int_{0^-}^{t} f(x)\,dx\right\} = \frac{F(s)}{s}$$

(a) find Zab

(b) Find poles and zeros of Zab



(a) find Zab



$$Z_2 = \frac{1}{sC_1} // (R_2 + sL_1) = \frac{\frac{1}{sC_1}\times(R_2 + sL_1)}{\frac{1}{sC_1} + R_2 + sL_1}$$

$$= \frac{R_2 + sL_1}{s^2 L_1 C_1 + sR_2C_1 + 1}$$

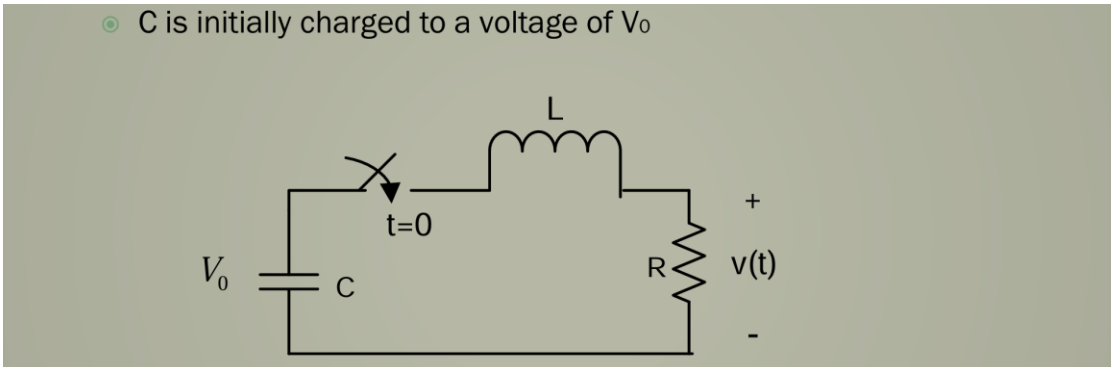
1. 它们之间的数学关系

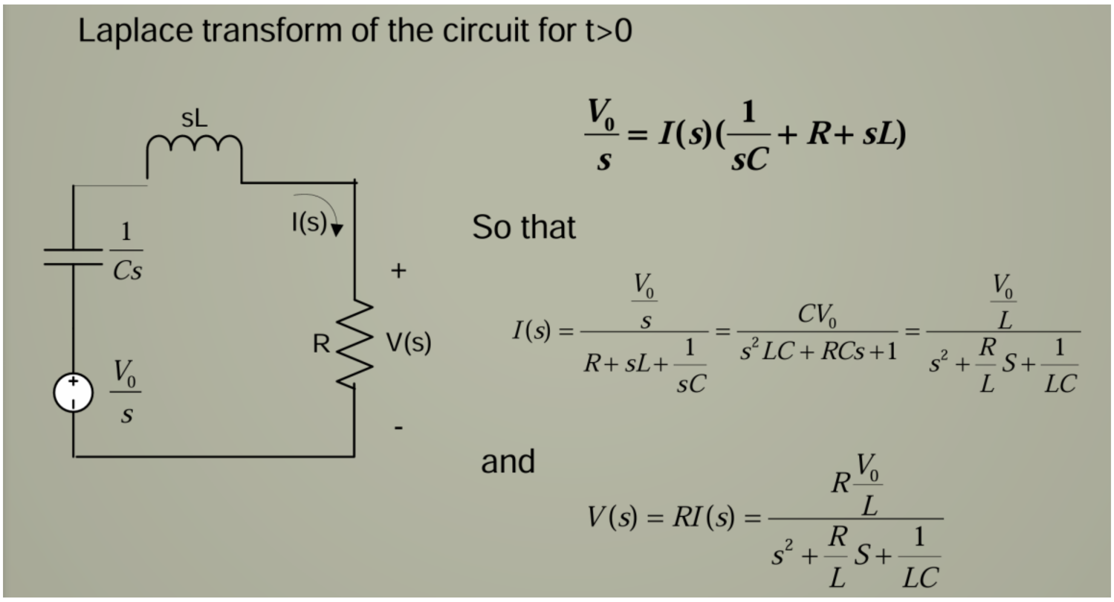
在拉普拉斯变换中，变量 s 是一个复数，通常写成：

$$s = \sigma + j\omega$$

其中 σ 代表衰减指数，而 $j\omega$ 代表角频率。

- **当你学 $X_C = \frac{1}{j\omega C}$ 时**：你研究的是**正弦稳态**。这时候假设电路已经通电很久了，波形是稳定的正弦波，没有衰减（即 $\sigma = 0$）。此时 s 直接变成了 $j\omega$。
- **当你学 $Z_C = \frac{1}{sC}$ 时**：你研究的是**全响应**。它不仅包括了稳态，还包括了开关合上那一刻的暂态（瞬态）过程（即 $\sigma \neq 0$ 的部分）。





Next to take the inverse transform

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0 = s^2 + 2\alpha s + \omega_o^2$$

ω_0 = resonant radian frequency = $\sqrt{\frac{1}{LC}}$

α = Neper frequency = $\frac{R}{2L}$

The poles will be the roots of the denominator polynomial

$$s^2 = \frac{-2\alpha \pm \sqrt{4\alpha^2 - 4\omega_0^2}}{2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -P_1; -P_2$$

- Depending upon the values of α and ω_o have 3 cases to consider:

(1) If $\alpha^2 > \omega_0^2 (R^2 > \frac{4L}{C})$ (over damped)

$-P_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}; -P_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$ (2 real distinct roots)

(2) If $\alpha^2 = \omega_0^2 (R^2 = \frac{4L}{C})$ (critical damped)

$-P_1 = -P_2 = -\alpha$ (2 real repeated roots)

(3) If $\alpha^2 < \omega_0^2 (R^2 < \frac{4L}{C})$ (under damped)

$-P_1 = -\alpha + j\beta; -P_2 = -\alpha - j\beta$

where $\beta = \sqrt{\omega_0^2 - \alpha^2}$ (2 complex conjugate roots)

Case 1: Over Damped

Poles are $-P_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}; -P_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$

$$V(s) = \frac{\frac{R}{L}V_0}{s^2 + \frac{R}{L}s + \frac{1}{LC}} = \frac{\frac{R}{L}V_0}{(s+P_1)(s+P_2)}$$

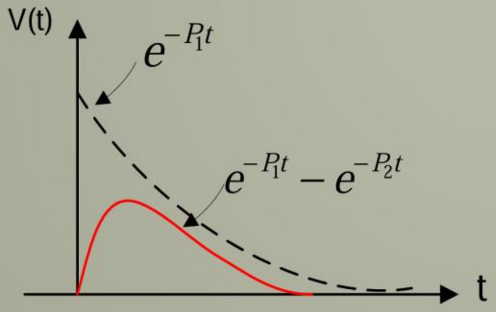
Perform a partial fraction expansion

$$V(s) = \frac{\frac{1}{2\sqrt{\alpha^2 - \omega_0^2}}\frac{R}{L}V_0}{s + \alpha - \sqrt{\alpha^2 - \omega_0^2}} + \frac{\frac{-1}{2\sqrt{\alpha^2 - \omega_0^2}}\frac{R}{L}V_0}{s + \alpha + \sqrt{\alpha^2 - \omega_0^2}}$$

Case 1: Over Damped

The time-domain response

$$v(t) = \frac{R}{2L} V_0 \frac{1}{\sqrt{\alpha^2 - \omega_0^2}} \left(e^{-(\alpha - \sqrt{\alpha^2 - \omega_0^2})t} - e^{-(\alpha + \sqrt{\alpha^2 - \omega_0^2})t}\right) u(t)$$



$\boldsymbol{P_1 < P_2; so}$

$e^{-P_2 t}$ **decay faster than** $e^{-P_1 t}$

$$v(t) = \frac{V_0 \alpha}{\sqrt{\alpha^2 - \omega_0^2}} (e^{-P_1 t} - e^{-P_2 t}) u(t)$$

P_1 falling time constant

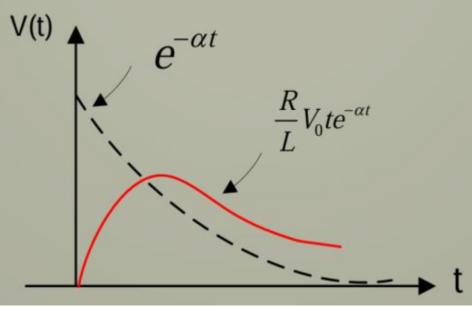
P_2 rising time constant

Case 2: Critically Damped

Poles are $-P_1 = -P_2 = -\alpha$

$$V(s) = \frac{\frac{R}{L} V_0}{(\alpha + s)^2} \qquad v(t) = (A + Bt) e^{-\alpha t} u(t)$$

Where A=1 B= $\frac{R}{L} V_0$



Case 3: Under Damped

$$v(t) = V(s) = \frac{\frac{R}{L}V_0}{(s+\alpha-j\beta)(s+\alpha+j\beta)}$$

$$V(s) = \frac{\frac{1}{j2\beta}\frac{R}{L}V_0}{(s+\alpha-j\beta)} + \frac{\frac{-1}{j2\beta}\frac{R}{L}V_0}{(s+\alpha+j\beta)}$$

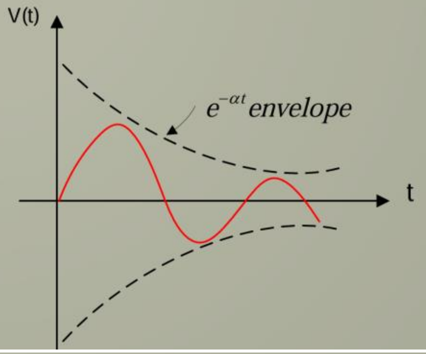
The inverse Laplace transform of this is:

$$v(t) = V_0\frac{\alpha}{\beta}\frac{1}{j}[e^{-\alpha t}e^{j\beta t} - e^{-\alpha t}e^{-j\beta t}]u(t)$$

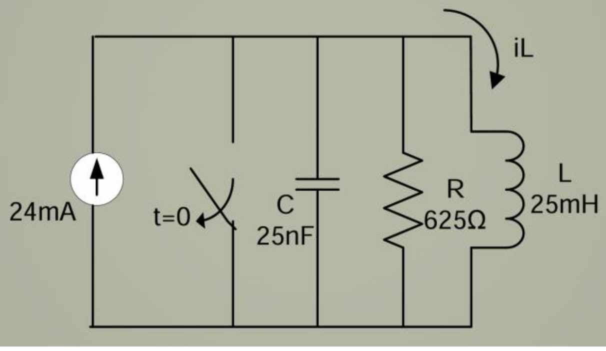
or using

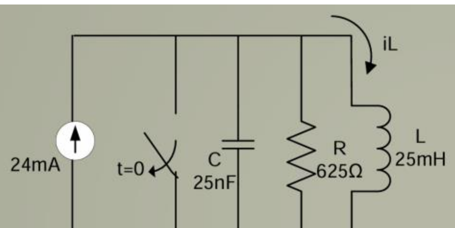
$$\sin\beta t = \frac{e^{j\beta t} - e^{-j\beta t}}{2j}$$

$$v(t) = V_0\frac{2\alpha}{\beta}e^{-\alpha t}\sin(\beta t)u(t)$$



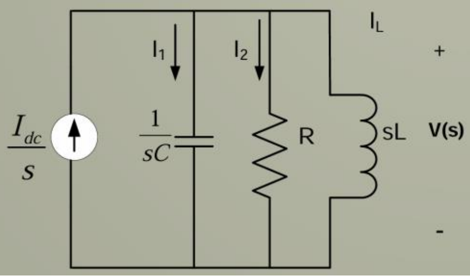
- The step response of a parallel circuit
- Find: $i_L(t)$





$$I_L = \frac{V(s)}{sL}$$

S-domain



$$\frac{I_{dc}}{s} = I_1 + I_2 + I_L = \frac{V(s)}{1/sC} + \frac{V(s)}{R} + \frac{V(s)}{sL}$$

$$V(s) = \frac{\frac{I_{dc}}{C}}{s^2 + (\frac{1}{RC})s + \frac{1}{LC}}$$

$$I_L = V(s)\frac{1}{sL} = \frac{\frac{I_{dc}}{LC}}{s(s^2 + (\frac{1}{RC})s + \frac{1}{LC})}$$

- The step response of a parallel circuit

$$I_L = V(s)\frac{1}{sL} = \frac{\frac{I_{dc}}{LC}}{s(s^2 + (\frac{1}{RC})s + \frac{1}{LC})}$$

Substituting the numerical values of R, C, L, Idc

$$I_L = \frac{384x10^5}{s(s^2 + 64000s + 16x10^8)}$$

$$Roots: s = 0; s = -32000 \pm j24000$$

$$I_L = \frac{384x10^5}{s(s + 32000 - j24000)(s + 32000 + j24000)}$$

- The step response of a parallel circuit

Perform a partial fraction expansion

$$I_L = \frac{k_1}{s} + \frac{k_2}{s + 32000 + j24000} + \frac{k_2^*}{s + 32000 - j24000}$$

$$k_1 = 24x10^{-3}; k_2 = 20x10^{-3}\angle 126.87^o$$

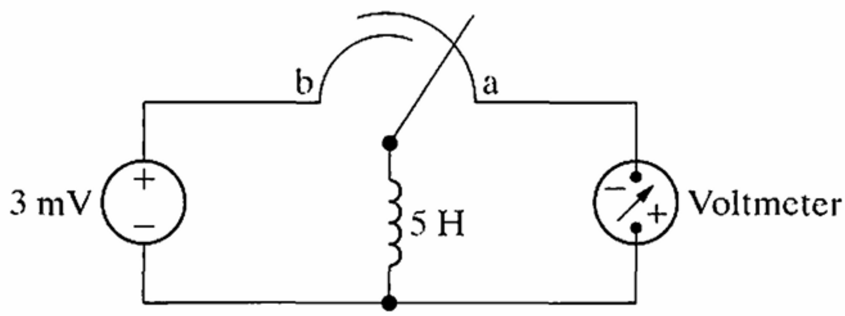
$$i_L = [24 + 40e^{-32000t}\cos(24000t + 126.87^o)]u(t)\,mA$$

Check Final value of $I_L; t \to \infty$ the inductor will be short circuit $I_L = 24mA(\lim_{s\to 0})$ $i(t \to \infty) = 24mA$

6.13 Initially there was no energy stored in the 5 H inductor in the circuit in Fig. P6.13 when it was placed across the terminals of the voltmeter. At

$t = 0$ the inductor was switched instantaneously to position b where it remained for 1.6 s before returning instantaneously to position a. The d'Arsonval voltmeter has a full-scale reading of 20 V and a sensitivity of 1000 Ω/V. What will the reading of the voltmeter be at the instant the switch returns to position a if the inertia of the d'Arsonval movement is negligible?

Figure P6.13



P 6.13 For $0 \leq t \leq 1.6\,\text{s}$:

$$i_L = \frac{1}{5}\int_0^t 3 \times 10^{-3}\,dx + 0 = 0.6 \times 10^{-3}t$$

$$i_L(1.6\,\text{s}) = (0.6 \times 10^{-3})(1.6) = 0.96\,\text{mA}$$

$$R_m = (20)(1000) = 20\,\text{k}\Omega$$

$$v_m(1.6\,\text{s}) = (0.96 \times 10^{-3})(20 \times 10^3) = 19.2\,\text{V}$$

6.23 The three inductors in the circuit in Fig. P6.23 are connected across the terminals of a black box at $t = 0$. The resulting voltage for $t > 0$ is known to be

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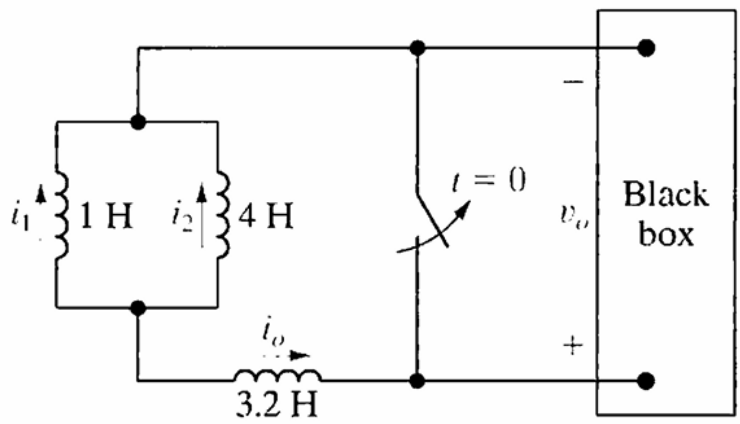
$$v_o = 2000e^{-100t}\ \text{V}.$$

If $i_1(0) = -6$ A and $i_2(0) = 1$ A, find

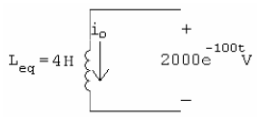
a) $i_o(0)$;
b) $i_o(t),\ t \geq 0$;
c) $i_1(t),\ t \geq 0$;
d) $i_2(t),\ t \geq 0$;
e) the initial energy stored in the three inductors;
f) the total energy delivered to the black box; and
g) the energy trapped in the ideal inductors.

Figure P6.23



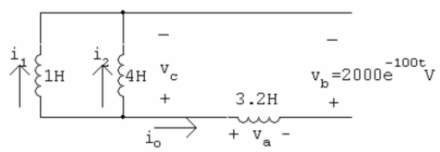
P 6.23 [a] $i_o(0) = -i_1(0) - i_2(0) = 6 - 1 = 5\,\text{A}$

[b]



$$\begin{aligned} i_o &= -\frac{1}{4}\int_0^t 2000e^{-100x}\,dx + 5 = -500\,\frac{e^{-100x}}{-100}\bigg|_0^t + 5 \\ &= 5(e^{-100t} - 1) + 5 = 5e^{-100t}\,\text{A}, \qquad t \geq 0 \end{aligned}$$

[c]



$$v_a = 3.2(-500e^{-100t}) = -1600e^{-100t}\,\text{V}$$

$$\begin{aligned} v_c &= v_a + v_b = -1600e^{-100t} + 2000e^{-100t} \\ &= 400e^{-100t}\,\text{V} \\ i_1 &= \frac{1}{1}\int_0^t 400e^{-100x}\,dx - 6 \\ &= -4e^{-100t} + 4 - 6 \\ i_1 &= -4e^{-100t} - 2\,\text{A} \qquad t \geq 0 \end{aligned}$$

[d] $$\begin{aligned} i_2 &= \frac{1}{4}\int_0^t 400e^{-100x}\,dx + 1 \\ &= -e^{-100t} + 2\,\text{A}, \qquad t \geq 0 \end{aligned}$$

[e] $w(0) = \frac{1}{2}(1)(6)^2 + \frac{1}{2}(4)(1)^2 + \frac{1}{2}(3.2)(5)^2 = 60\,\text{J}$

[f] $w_{\text{del}} = \frac{1}{2}(4)(5)^2 = 50\,\text{J}$

[g] $w_{\text{trapped}} = 60 - 50 = 10\,\text{J}$

or $\quad w_{\text{trapped}} = \frac{1}{2}(1)(2)^2 + \frac{1}{2}(4)(2)^2 + 10\,\text{J (check)}$

计算电感的能量的时候，必须要分开独立计算。

常规的理解方式，可以先计算出初始的能量，然后计算出最终的能量，传递的能量通过二者相减得到

可以通过等效的元件，得出传递的能量

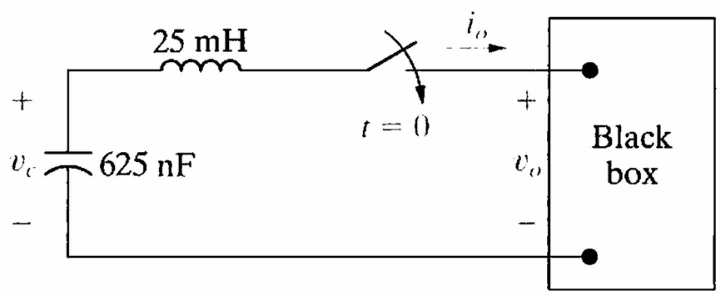
先通过等效元件，得出总的电流，在对每一个小的元件进行积分，得到对应的电流

6.35 At $t = 0$, a series-connected capacitor and inductor are placed across the terminals of a black box, as shown in Fig. P6.35. For $t > 0$, it is known that

$$i_o = 1.5e^{-16{,}000t} - 0.5e^{-4000t}\,\text{A}.$$

If $v_c(0) = -50$ V find v_o for $t \geq 0$.

Figure P6.35



$$
\begin{aligned}
\text{P 6.35}\qquad v_c &= -\frac{1}{0.625\times10^{-6}}\left(\int_0^t 1.5e^{-16{,}000x}\,dx-\int_0^t 0.5e^{-4000x}\,dx\right)-50\\
&= 150(e^{-16{,}000t}-1)-200(e^{-4000t}-1)-50\\
&= 150e^{-16{,}000t}-200e^{-4000t}\ \text{V}\\
v_L &= 25\times10^{-3}\frac{di_o}{dt}\\
&= 25\times10^{-3}(-24{,}000e^{-16{,}000t}+2000e^{-4000t})\\
&= -600e^{-16{,}000t}+50e^{-4000t}\ \text{V}\\
v_o &= v_c-v_L\\
&= (150e^{-16{,}000t}-200e^{-4000t})-(-600e^{-16{,}000t}+50e^{-4000t})\\
&= 750e^{-16{,}000t}-250e^{-4000t}\ \text{V},\ t>0
\end{aligned}
$$

注意电感和电容的符号，可以通过楞次定律检查符号

7.3 The switch in the circuit in Fig. P7.3 has been open for a long time. At $t = 0$ the switch is closed.

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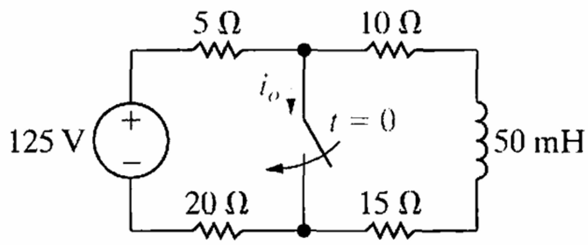
a) Determine $i_o(0^+)$ and $i_o(\infty)$.

b) Determine $i_o(t)$ for $t \geq 0^+$.

c) How many milliseconds after the switch has been closed will the current in the switch equal 3 A?

Figure P7.3



P 7.3 [a] $i_L(0)=\dfrac{125}{50}=2.5\,\text{A}$

$i_o(0^+)=\dfrac{125}{25}-2.5=5-2.5=2.5\,\text{A}$

$i_o(\infty)=\dfrac{125}{25}=5\,\text{A}$

[b] $i_L=2.5e^{-t/\tau};\qquad \tau=\dfrac{L}{R}=\dfrac{50\times10^{-3}}{25}=2\,\text{ms}$

$i_L=2.5e^{-500t}\,\text{A}$

$i_o=5-i_L=5-2.5e^{-500t}\,\text{A},\qquad t\geq0^+$

[c] $5-2.5e^{-500t}=3$

$2=2.5e^{-500t}$

$e^{500t}=1.25\qquad \therefore\ t=446.29\,\mu\text{s}$

将左右分成两个独立的电路，当 t=0 时，左侧电路对 i 的作用为 5 右侧电路为反向的 2.5

时间常数主要时观察放电过程中，哪些电阻对其有影响，电流只在右侧的环中，所以电阻为 25

注意：电阻中的电流可以瞬间变化

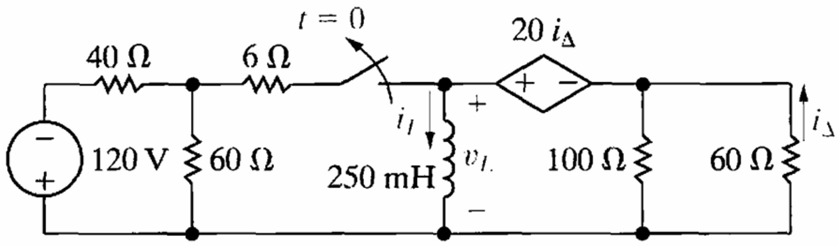
7.14 The switch in Fig. P7.14 has been closed for a long time before opening at $t = 0$. Find

a) $i_L(t)$, $t \geq 0$.

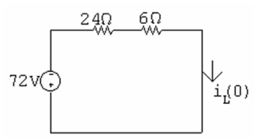
b) $v_L(t)$, $t \geq 0^+$.

c) $i_\Delta(t)$, $t \geq 0^+$.

Figure P7.14

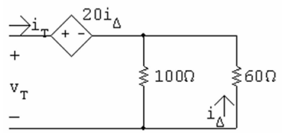


P 7.14 [a] $t < 0$:



$$i_L(0) = -\frac{72}{24+6} = -2.4\,\text{A}$$

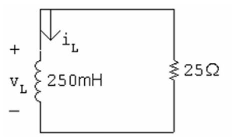
$t > 0$:



$$i_\Delta = -\frac{100}{160}i_T = -\frac{5}{8}i_T$$

$$v_T = 20i_\Delta + i_T\frac{(100)(60)}{160} = -12.5i_T + 37.5i_T$$

$$\frac{v_T}{i_T} = R_{\text{Th}} = -12.5 + 37.5 = 25\,\Omega$$



$$\tau = \frac{L}{R} = \frac{250 \times 10^{-3}}{25} \qquad \frac{1}{\tau} = 100$$

$$i_L = -2.4e^{-100t}\,\text{A}, \qquad t \geq 0$$

[b] $v_L = 250 \times 10^{-3}(240e^{-100t}) = 60e^{-100t}\,\text{V}, \quad t \geq 0^+$

[c] $i_\Delta = 0.625i_L = -1.5e^{-100t}\,\text{A} \qquad t \geq 0^+$

7.15 What percentage of the initial energy stored in the inductor in the circuit in Fig. P7.14 is dissipated by the 60 Ω resistor?

P 7.15 $w(0) = \frac{1}{2}(250 \times 10^{-3})(-2.4)^2 = 720\,\text{mJ}$

$$p_{60\Omega} = 60(-1.5e^{-100t})^2 = 135e^{-200t}\ \text{W}$$

$$w_{60\Omega} = \int_0^\infty 135e^{-200t}\,dt = 135\frac{e^{-200t}}{-200}\Big|_0^\infty = 675\,\text{mJ}$$

$$\%\ \text{dissipated} = \frac{675}{720}(100) = 93.75\%$$

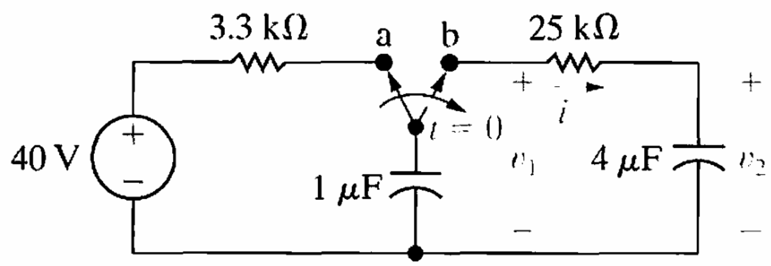
7.23 The switch in the circuit in Fig. P7.23 has been in position a for a long time and $v_2 = 0$ V. At $t = 0$, the switch is thrown to position b. Calculate

a) i, v_1, and v_2 for $t \geq 0^+$.

b) the energy stored in the capacitor at $t = 0$.

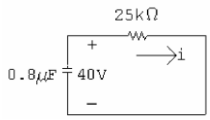
c) the energy trapped in the circuit and the total energy dissipated in the 25 kΩ resistor if the switch remains in position b indefinitely.

Figure P7.23



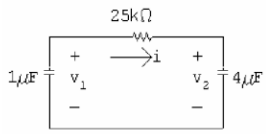
P 7.23 [a] $v_1(0^-) = v_1(0^+) = 40\,\text{V}$ $\qquad v_2(0^+) = 0$

$$C_{eq} = (1)(4)/5 = 0.8\,\mu\text{F}$$



$$\tau = (25 \times 10^3)(0.8 \times 10^{-6}) = 20\text{ms}; \qquad \frac{1}{\tau} = 50$$

$$i = \frac{40}{25{,}000}e^{-50t} = 1.6e^{-50t}\,\text{mA}, \qquad t \geq 0^+$$



$$v_1 = \frac{-1}{10^{-6}}\int_0^t 1.6\times 10^{-3}e^{-50x}\,dx + 40 = 32e^{-50t} + 8\,\text{V}, \qquad t \geq 0$$

$$v_2 = \frac{1}{4\times 10^{-6}}\int_0^t 1.6\times 10^{-3}e^{-50x}\,dx + 0 = -8e^{-50t} + 8\,\text{V}, \qquad t \geq 0$$

[b] $w(0) = \frac{1}{2}(10^{-6})(40)^2 = 800\,\mu\text{J}$

[c] $w_{\text{trapped}} = \frac{1}{2}(10^{-6})(8)^2 + \frac{1}{2}(4\times 10^{-6})(8)^2 = 160\,\mu\text{J}.$

The energy dissipated by the 25 kΩ resistor is equal to the energy dissipated by the two capacitors; it is easier to calculate the energy dissipated by the capacitors:

$$w_{\text{diss}} = \frac{1}{2}(0.8\times 10^{-6})(40)^2 = 640\,\mu\text{J}.$$

Check: $w_{\text{trapped}} + w_{\text{diss}} = 160 + 640 = 800\,\mu\text{J}; \qquad w(0) = 800\,\mu\text{J}.$

7.41 The switch in the circuit in Fig. P7.41 has been open a long time before closing at $t = 0$. Find $v_o(t)$ for $t \geq 0^+$.

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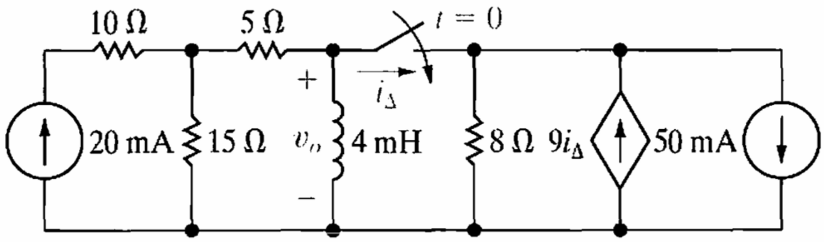
Figure P7.41

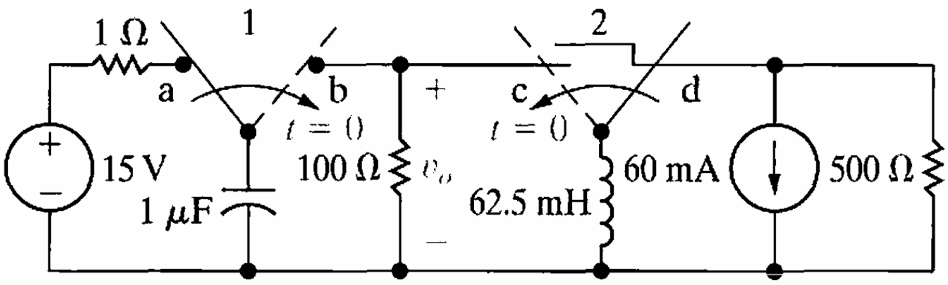


先对左右各进行一次戴维南等效，最后将左右两次一起等效成一个戴维南

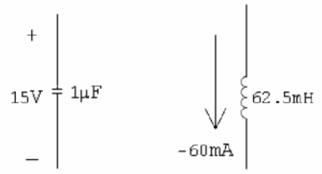
8.18 The two switches in the circuit seen in Fig. P8.18 operate synchronously. When switch 1 is in position a, switch 2 is in position d. When switch 1 moves to position b, switch 2 moves to position c. Switch 1 has been in position a for a long time. At $t = 0$, the switches move to their alternate positions. Find $v_o(t)$ for $t \geq 0$.

PSPICE
MULTISIM

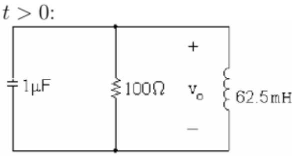
Figure P8.18



P 8.18 $t < 0$: $V_o = 15\,\text{V}$, $I_o = -60\,\text{mA}$



$t > 0$:



$$i_R(0) = \frac{15}{100} = 150\,\text{mA}; \qquad i_L(0) = -60\,\text{mA}$$

$$i_C(0) = -150 - (-60) = -90\,\text{mA}$$

$$\alpha = \frac{1}{2RC} = \frac{1}{2(100)(10^{-6})} = 5000\ \text{rad/s}$$

$$\omega_o^2 = \frac{1}{LC} = \frac{1}{(62.5 \times 10^{-3})(10^{-6})} = 16 \times 10^6$$

$$s_{1,2} = -5000 \pm \sqrt{25 \times 10^6 - 16 \times 10^6} = -5000 \pm 3000$$

$$s_1 = -2000\ \text{rad/s}; \qquad s_2 = -8000\ \text{rad/s}$$

$$\therefore \quad v_o = A_1 e^{-2000t} + A_2 e^{-8000t}$$

$$A_1 + A_2 = v_o(0) = 15$$

$$\frac{dv_o}{dt}(0) = -2000A_1 - 8000A_2 = \frac{-90 \times 10^{-3}}{10^{-6}} = -90{,}000$$

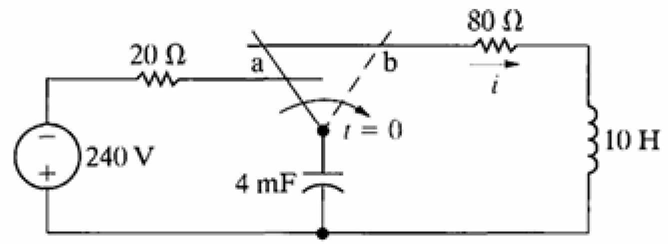
Solving, $A_1 = 5\,\text{V}$, $A_2 = 10\,\text{V}$

$$\therefore \quad v_o = 5e^{-2000t} + 10e^{-8000t}\ \text{V}, \qquad t \geq 0$$

第一步：将电路化简到最简形式
第二步：判断是并联还是串联 RLC
第三步：计算衰减系数与谐振频率，判断是哪种类型的振荡
第四步：求出特征根
第五步：带出 0 时刻，A_1+A_2=初始的电压或者电流
第六步：根据 KCL，列出等式（初始就有的电流之间带出，还有一个需要求一次导）

8.45 PSPICE MULTISIM The switch in the circuit shown in Fig. P8.45 has been in position a for a long time. At $t = 0$, the switch is moved instantaneously to position b. Find $i(t)$ for $t \geq 0$.

Figure P8.45



P 8.45 $\omega_o^2 = \dfrac{1}{LC} = \dfrac{1}{(10)(4\times 10^{-3})} = 25$

$\alpha = \dfrac{R}{2L} = \dfrac{80}{2(10)} = 4; \qquad \alpha^2 = 16$

$\alpha^2 < \omega_o^2 \quad \therefore \qquad \text{underdamped}$

$s_{1,2} = -4 \pm j\sqrt{9} = -4 \pm j3 \text{ rad/s}$

$i = B_1 e^{-4t}\cos 3t + B_2 e^{-4t}\sin 3t$

$i(0) = B_1 = -240/100 = -2.4\,\text{A}$

$\dfrac{di}{dt}(0) = 3B_2 - 4B_1 = 0$

$\therefore \quad B_2 = -3.2\,\text{A}$

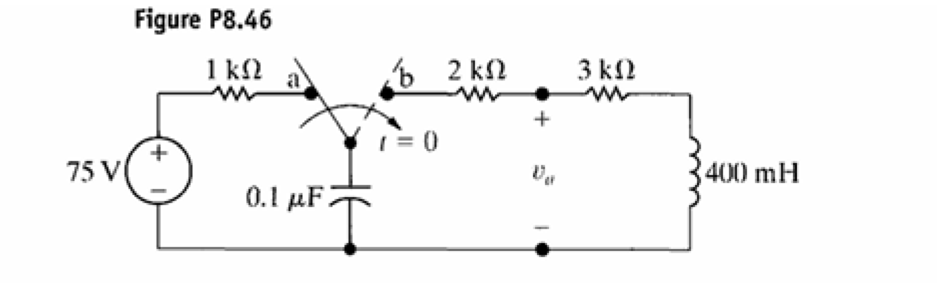
$i = -2.4e^{-4t}\cos 3t - 3.2\sin 3t\,\text{A}, \qquad t \geq 0$

8.46 PSPICE MULTISIM The switch in the circuit in Fig. P8.46 on the next page has been in position a for a long time. At $t = 0$, the switch moves instantaneously to position b.

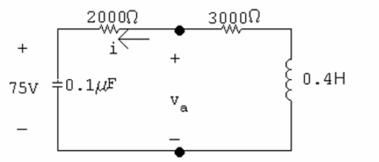
a) What is the initial value of v_a?

b) What is the initial value of dv_a/dt?

c) What is the numerical expression for $v_a(t)$ for $t \geq 0$?



P 8.46 [a] For $t > 0$:



Since $i(0^-) = i(0^+) = 0$

$v_a(0^+) = 75\,\text{V}$

[b] $v_a = 2000i + 10^7 \int_0^t i\,dx + 75$

$$\frac{dv_a}{dt} = 2000\frac{di}{dt} + 10^7 i$$

$$\frac{dv_a(0^+)}{dt} = 2000\frac{di(0^+)}{dt} + 10^7 i(0^+) = 2000\frac{di(0^+)}{dt}$$

$$-L\frac{di(0^+)}{dt} = 75$$

$$\frac{di(0^+)}{dt} = -2.5(75) = -187.5\,\text{A/s}$$

$$\therefore \quad \frac{dv_a(0^+)}{dt} = -375{,}000\,\text{V/s}$$

[c] $\alpha = \dfrac{R}{2L} = \dfrac{5000}{0.8} = 6250\,\text{rad/s}$

$$\omega_o^2 = \frac{1}{LC} = \frac{10^6}{(0.4)(0.1)} = 25 \times 10^6$$

$$s_{1,2} = -6250 \pm \sqrt{6250^2 - 25 \times 10^6} = -6250 \pm 3750\,\text{rad/s}$$

$$\therefore \quad s_1 = -2500\,\text{rad/s}; \qquad s_2 = -10{,}000\,\text{rad/s}$$

Overdamped:

$$v_a = A_1 e^{-2500t} + A_2 e^{-10{,}000t}$$

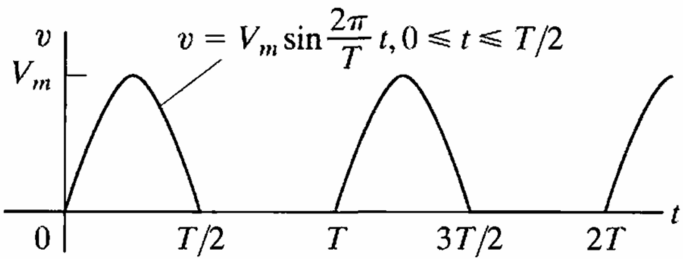
$$v_a(0) = A_1 + A_2 = 75\,\text{V}$$

$$\frac{dv_a(0)}{dt} = -2500A_1 - 10{,}000A_2 = -375{,}000; \qquad \therefore \quad A_1 = 50\,\text{V}, \qquad A_2 = 25\,\text{V}$$

$$v_a = 50e^{-2500t} + 25e^{-10{,}000t}\,\text{V}, \quad t \geq 0^+$$

9.8 Find the rms value of the half-wave rectified sinusoidal voltage shown.

Figure P9.8



P 9.8 $V_{\rm rms} = \sqrt{\frac{1}{T}\int_0^{T/2} V_m^2 \sin^2 \frac{2\pi}{T}t\, dt}$

$$\int_0^{T/2} V_m^2 \sin^2\left(\frac{2\pi}{T}t\right)\, dt = \frac{V_m^2}{2}\int_0^{T/2}\left(1 - \cos\frac{4\pi}{T}t\right)\, dt = \frac{V_m^2 T}{4}$$

Therefore $V_{\rm rms} = \sqrt{\frac{1}{T}\frac{V_m^2 T}{4}} = \frac{V_m}{2}$

9.9 The voltage applied to the circuit shown in Fig. 9.5 at $t = 0$ is $20\cos(800t + 25°)$ V. The circuit resistance is 80 Ω and the initial current in the 75 mH inductor is zero.

a) Find $i(t)$ for $t \geq 0$.

b) Write the expressions for the transient and steady-state components of $i(t)$.

c) Find the numerical value of i after the switch has been closed for 1.875 ms.

d) What are the maximum amplitude, frequency (in radians per second), and phase angle of the steady-state current?

e) By how many degrees are the voltage and the steady-state current out of phase?

P 9.9 [a] The numerical values of the terms in Eq. 9.8 are

$V_m = 20, \qquad R/L = 1066.67, \qquad \omega L = 60$

$\sqrt{R^2 + \omega^2 L^2} = 100$

$\phi = 25°, \qquad \theta = \tan^{-1} 60/80, \qquad \theta = 36.87°$

Substitute these values into Equation 9.9:

$i = \left[-195.72e^{-1066.67t} + 200\cos(800t - 11.87°)\right]\ \text{mA}, \qquad t \geq 0$

[b] Transient component $= -195.72e^{-1066.67t}$ mA
Steady-state component $= 200\cos(800t - 11.87°)$ mA

[c] By direct substitution into Eq 9.9 in part (a), $i(1.875\,\text{ms}) = 28.39\,\text{mA}$

[d] 200 mA, 800 rad/s, −11.87°

[e] The current lags the voltage by 36.87°.

注意 θ 的表达式，是$\theta = \arctan(\omega L/R)$

9.11 Use the concept of the phasor to combine the following sinusoidal functions into a single trigonometric expression:

a) $y = 50\cos(500t + 60°) + 100\cos(500t - 30°)$,

b) $y = 200\cos(377t + 50°) - 100\sin(377t + 150°)$,

P 9.11 [a] $\mathbf{Y} = 50\underline{/60°} + 100\underline{/-30°} = 111.8\underline{/-3.43°}$

$y = 111.8\cos(500t - 3.43°)$

[b] $\mathbf{Y} = 200\underline{/50°} - 100\underline{/60°} = 102.99\underline{/40.29°}$

$y = 102.99\cos(377t + 40.29°)$

[c] $\mathbf{Y} = 80\underline{/30°} - 100\underline{/-225°} + 50\underline{/-90°} = 161.59\underline{/-29.96°}$

$y = 161.59\cos(100t - 29.96°)$

先将 cos 进行相量转换，然后在换成复数，复数加减之后，再写出对应的相量，最后转回 cos

9.12 The expressions for the steady-state voltage and current at the terminals of the circuit seen in Fig. P9.12 are

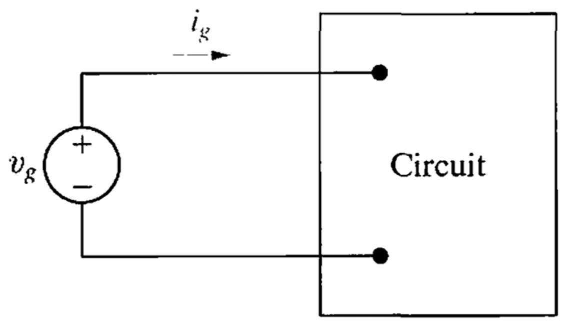
$$v_g = 300\cos(5000\pi t + 78°)\ \text{V},$$

$$i_g = 6\sin(5000\pi t + 123°)\ \text{A}$$

a) What is the impedance seen by the source?

b) By how many microseconds is the current out of phase with the voltage?

Figure P9.12



P 9.12 [a] $\mathbf{V}_g = 300\underline{/78^\circ}$; $\quad \mathbf{I}_g = 6\underline{/33^\circ}$

$$\therefore \quad Z = \frac{\mathbf{V}_g}{\mathbf{I}_g} = \frac{300\underline{/78^\circ}}{6\underline{/33^\circ}} = 50\underline{/45^\circ}\,\Omega$$

[b] i_g lags v_g by 45°:

$2\pi f = 5000\pi$; $\quad f = 2500\,\text{Hz}$; $\quad T = 1/f = 400\,\mu\text{s}$

$$\therefore \quad i_g \text{ lags } v_g \text{ by } \frac{45^\circ}{360^\circ}(400\,\mu\text{s}) = 50\,\mu\text{s}$$

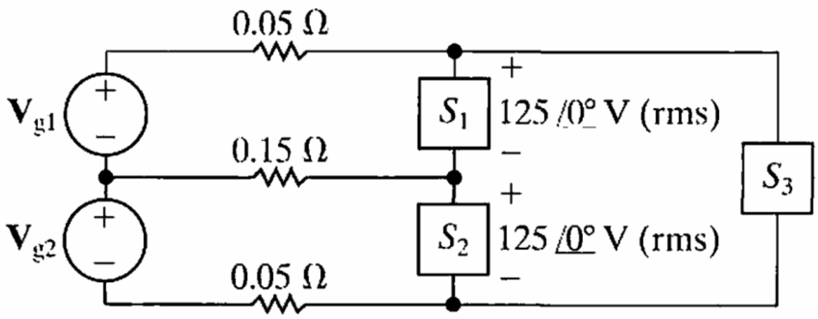
阻抗直接使用相量来计算，不用再转会复数

10.24 The three loads in the circuit shown in Fig. P10.24 are $S_1 = 5 + j1.25$ kVA, $S_2 = 6.25 + j2.5$ kVA, and $S_3 = 8 + j0$ kVA.

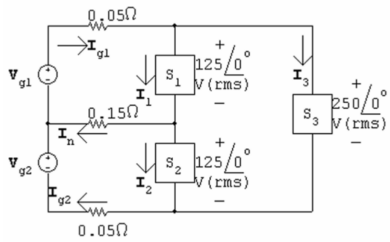
a) Calculate the complex power associated with each voltage source, $\mathbf{V}_{g1}$ and $\mathbf{V}_{g2}$.

b) Verify that the total real and reactive power delivered by the sources equals the total real and reactive power absorbed by the network.

Figure P10.24



P 10.24 [**a**]



$$\mathbf{I}_1 = \frac{5000 - j1250}{125} = 40 - j10\,\text{A (rms)}$$

$$\mathbf{I}_2 = \frac{6250 - j2500}{125} = 50 - j20\,\text{A (rms)}$$

$$\mathbf{I}_3 = \frac{8000 + j0}{250} = 32 + j0\,\text{A (rms)}$$

$$\therefore \quad \mathbf{I}_{g1} = 72 - j10\,\text{A (rms)}$$

$$\mathbf{I}_n = \mathbf{I}_1 - \mathbf{I}_2 = -10 + j10\,\text{A (rms)}$$

$$\mathbf{I}_{g2} = 82 - j20\,\text{A}$$

$$\mathbf{V}_{g1} = 0.05\mathbf{I}_{g1} + 125 + j0 + 0.15\mathbf{I}_n = 127.1 - j1\,\text{V(rms)}$$

$$\mathbf{V}_{g2} = -0.15\mathbf{I}_n + 125 + j0 + 0.05\mathbf{I}_{g2} = 130.6 - j2.5\,\text{V(rms)}$$

$$S_{g1} = -[(127.1 - j1)(72 + j10)] = -[9141.2 + j1343]\,\text{VA}$$

$$S_{g2} = -[(130.6 - j2.5)(82 + j20)] = -[10{,}759.2 + j2407]\,\text{VA}$$

Note: Both sources are delivering average power and magnetizing VAR to the circuit.

[**b**] $P_{0.05} = |\mathbf{I}_{g1}|^2(0.05) = 264.2\,\text{W}$

$$P_{0.15} = |\mathbf{I}_n|^2(0.15) = 30\,\text{W}$$

$$P_{0.05} = |\mathbf{I}_{g2}|^2(0.05) = 356.2\,\text{W}$$

$$\sum P_{\text{dis}} = 264.2 + 30 + 356.2 + 5000 + 8000 + 6250 = 19{,}900.4\,\text{W}$$

$$\sum P_{\text{dev}} = 9141.2 + 10{,}759.2 = 19{,}900.4\,\text{W} = \sum P_{\text{dis}}$$

$$\sum Q_{\text{abs}} = 1250 + 2500 = 3750\,\text{VAR}$$

$$\sum Q_{\text{del}} = 1343 + 2407 = 3750\,\text{VAR} = \sum Q_{\text{abs}}$$

注意有些地方要使用共轭。正常计算 I*=S/V

10.50 The variable resistor R_o in the circuit shown in Fig. P10.50 is adjusted until maximum average power is delivered to R_o.

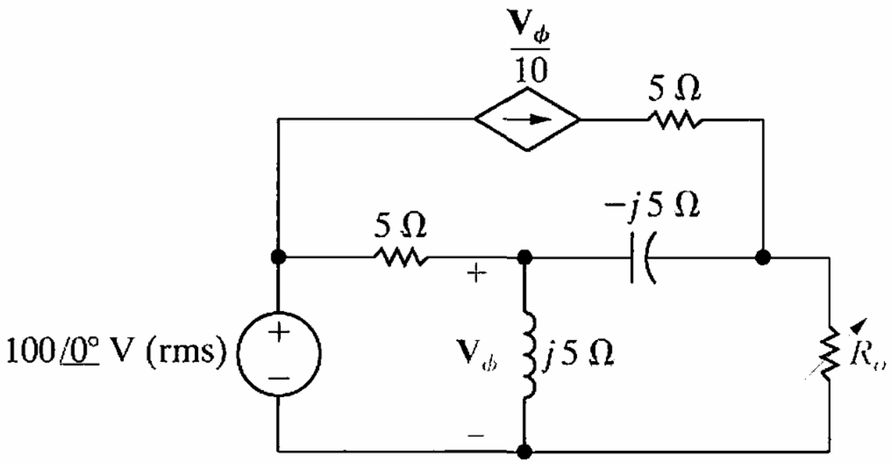
a) What is the value of R_o in ohms?

b) Calculate the average power delivered to R_o.

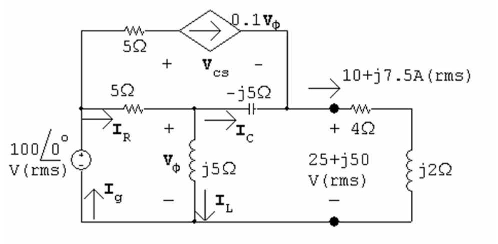
c) If R_o is replaced with a variable impedance Z_o, what is the maximum average power that can be delivered to Z_o?

d) In (c), what percentage of the circuit's developed power is delivered to the load Z_o?

Figure P10.50



[d]



$$\frac{\mathbf{V}_\phi - 100}{5} + \frac{\mathbf{V}_\phi}{j5} + \frac{\mathbf{V}_o - (25 + j50)}{-j5} = 0$$

$\mathbf{V}_\phi = 50 + j25\,\mathrm{V}$ (rms)

$0.1\mathbf{V}_\phi = 5 + j2.5\,\mathrm{V}$ (rms)

$5 + j2.5 + \mathbf{I}_C = 10 + j7.5$

$\mathbf{I}_C = 5 + j5\,\mathrm{A}$ (rms)

$\mathbf{I}_L = \dfrac{\mathbf{V}_\phi}{j5} = 5 - j10\,\mathrm{A}$ (rms)

$\mathbf{I}_R = \mathbf{I}_C + \mathbf{I}_L = 10 - j5\,\mathrm{A}$ (rms)

$\mathbf{I}_g = \mathbf{I}_R + 0.1\mathbf{V}_\phi = 15 - j2.5\,\mathrm{A}$ (rms)

$S_g = -100\mathbf{I}_g^* = -1500 - j250\,\mathrm{VA}$

$100 = 5(5 + j2.5) + \mathbf{V}_{\mathrm{cs}} + 25 + j50 \quad \therefore \quad \mathbf{V}_{\mathrm{cs}} = 50 - j62.5\,\mathrm{V}$ (rms)

$S_{\mathrm{cs}} = (50 - j62.5)(5 - j2.5) = 93.75 - j437.5\,\mathrm{VA}$

Thus,

$\sum P_{\mathrm{dev}} = 1500$

% delivered to $R_o = \dfrac{625}{1500}(100) = 41.67\%$

注意，受控电流源是在消耗功率

12.25 Find the Laplace transform for (a) and (b).

a) $f(t) = \dfrac{d}{dt}(e^{-at}\sin \omega t)$.

b) $f(t) = \displaystyle\int_{0^-}^{t} e^{-ax}\cos \omega x \, dx$.

c) Verify the results obtained in (a) and (b) by first carrying out the indicated mathematical operation and then finding the Laplace transform.

P 12.25 [a] $f_1(t) = e^{-at}\sin\omega t; \qquad F_1(s) = \frac{\omega}{(s+a)^2+\omega^2}$

$$F(s) = sF_1(s) - f_1(0^-) = \frac{s\omega}{(s+a)^2+\omega^2} - 0$$

[b] $f_1(t) = e^{-at}\cos\omega t; \qquad F_1(s) = \frac{s+a}{(s+a)^2+\omega^2}$

$$F(s) = \frac{F_1(s)}{s} = \frac{s+a}{s[(s+a)^2+\omega^2]}$$

[c] $\frac{d}{dt}[e^{-at}\sin\omega t] = \omega e^{-at}\cos\omega t - ae^{-at}\sin\omega t$

Therefore $F(s) = \frac{\omega(s+a)-\omega a}{(s+a)^2+\omega^2} = \frac{\omega s}{(s+a)^2+\omega^2}$

$$\int_{0^-}^{t} e^{-ax}\cos\omega x\,dx = \frac{-ae^{-at}\cos\omega t + \omega e^{-at}\sin\omega t + a}{a^2+\omega^2}$$

Therefore

$$F(s) = \frac{1}{a^2+\omega^2}\left[\frac{-a(s+a)}{(s+a)^2+\omega^2} + \frac{\omega^2}{(s+a)^2+\omega^2} + \frac{a}{s}\right]$$

$$= \frac{s+a}{s[(s+a)^2+\omega^2]}$$

14.25 For the bandpass filter shown in Fig. P14.25, calculate the following: (a) f_o; (b) Q; (c) f_{c1}; (d) f_{c2}; and (e) β.

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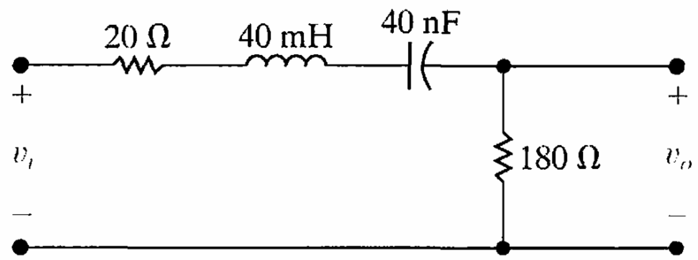
Figure P14.25



P 14.25 **[a]** $\omega_o^2 = \dfrac{1}{LC} = \dfrac{1}{(40\times10^{-3})(40\times10^{-9})} = 625\times10^6$

$$\omega_o = 25\times10^3 \text{ rad/s} = 25\,\text{krad/s}$$

$$f_o = \frac{25{,}000}{2\pi} = 3978.87\,\text{Hz}$$

[b] $Q = \dfrac{\omega_o L}{R+R_i} = \dfrac{(25\times10^3)(40\times10^{-3})}{200} = 5$

[c] $\omega_{c1} = \omega_o\left[-\dfrac{1}{2Q} + \sqrt{1+\left(\dfrac{1}{2Q}\right)^2}\right] = 25{,}000\left[-\dfrac{1}{10} + \sqrt{1+\dfrac{1}{100}}\right]$

$$= 22.62\,\text{krad/s} \quad \text{or} \quad 3.60\,\text{kHz}$$

[d] $w_{c2} = \omega_o\left[\dfrac{1}{2Q} + \sqrt{1+\left(\dfrac{1}{2Q}\right)^2}\right] = 25{,}000\left[\dfrac{1}{10} + \sqrt{1+\dfrac{1}{100}}\right]$

$$= 27.62\,\text{krad/s} \quad \text{or} \quad 4.4\,\text{kHz}$$

[e] $\beta = \omega_{c2} - \omega_{c1} = 27.62 - 22.62 = 5\,\text{krad/s}$

or

$$\beta = \frac{\omega_o}{Q} = \frac{25{,}000}{5} = 5\,\text{krad/s} \quad \text{or} \quad 795.77\,\text{Hz}$$

特别注意β的算法